

EE 210L: Digital Circuits and Logic Design Lab

Lab 3: Minterms and SOP Implementation

Name: _____ Date: _____

Partner(s): _____ Section: _____

Record logic levels only, 0 or 1. Write a variable plain where its row has a 1 and complemented where it has a 0.

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: Function G — Built Straight from the Minterms

Canonical SOP $G =$ _____

#	A	B	Predicted G	Measured G	74LS86 output
1	0	0			
2	0	1			
3	1	0			
4	1	1			

Which single gate did you build out of five? _____

Part 5.3: Function M — The Majority Voter

Minterm list $M = \sum m($ _____ $)$

Simplified expression $M =$ _____

#	A	B	C	Predicted M	Measured M
1	0	0	0		
2	0	0	1		
3	0	1	0		
4	0	1	1		
5	1	0	0		
6	1	0	1		
7	1	1	0		
8	1	1	1		

Does the measured column match the specification — output 1 exactly when two or three inputs are 1?

Part 5.4: Function F — A Variable That Disappears

Simplified expression $F =$ _____

#	A	B	C	Predicted F	Measured F
1	0	0	0		
2	0	0	1		
3	0	1	0		
4	0	1	1		
5	1	0	0		
6	1	0	1		
7	1	1	0		
8	1	1	1		

What does toggling C do to the output? Explain how six minterms that all contain C produced a function that does not depend on C.

Part 5.5: Cost of the Canonical Form

Count 2-input gates. An n -input gate costs $n - 1$ of them. Count inverters separately.

Function	Version	AND	OR	NOT	Total gates	Packages
M (majority)	Canonical					
M (majority)	Simplified					
F	Canonical					
F	Simplified					

Gates the canonical F would have needed: _____ in _____ packages

Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: why the canonical SOP works on any function without requiring you to be clever, what it costs you to skip the simplification step, and whether the circuit you built for G would have been obvious to you from the truth table alone.
